



NORTHSTAR DIGITAL SERVICES

Issue & Vulnerability Tracker

CLIENT RUN GUIDE

A step-by-step Windows guide for starting the MySQL database, Flask REST API, and Streamlit dashboard on a client computer.

Application	Northstar Risk Operations
Frontend	Streamlit - http://localhost:8501
Backend	Flask API - http://127.0.0.1:5000
Database	MySQL 8.4 in Docker

1. What the client needs

Install the following software before running the application. Administrator permission may be needed during installation.

Software	Purpose	Check command
Git	Download or clone the source code	git --version
Python 3.10+	Run Flask, Streamlit, and tests	python --version
Docker Desktop	Run the MySQL 8.4 database	docker --version

Repository access

If the GitHub repository is private, the client must be invited as a collaborator or receive an approved ZIP export. Never share a GitHub password or access token.

Option A - clone with Git

Open PowerShell in the folder where the application should be stored, then run:

```
git clone https://github.com/marshmannoor/Advance_Programming-CA2.git
cd Advance_Programming-CA2
```

Option B - use a ZIP

Download the repository ZIP, extract it, open the extracted folder, right-click an empty area, and choose **Open in Terminal**.

Confirm the project files

The folder should contain run.py, streamlit_app.py, requirements.txt, docker-compose.yml, app, tests, and README.md.

2. First-time setup

Step 1 - create a Ppython virtual environment

```
python -m venv .venv
.\.venv\Scripts\Activate.ps1
```

If PowerShell blocks activation, run the following once in the same terminal, then activate again:

```
Set-ExecutionPolicy -Scope Process -ExecutionPolicy Bypass
```

Step 2 - install Ppython packages

```
python -m pip install --upgrade pip
pip install -r requirements.txt
```

Step 3 - start Docker Desktop

Open Docker Desktop and wait until it reports that the engine is running. The first start can take a minute and may request WSL setup.

Step 4 - start MvSQL

```
docker compose up -d mysql
docker compose ps
```

The northstar-mysql service should show healthy. MySQL listens locally on port 3306.

Step 5 - create the database tables

```
python -m flask --app run.py init-db
```

Expected message: **Initialized the MySQL database.** This command is safe to run again because the tables use CREATE TABLE IF NOT EXISTS.

3. Start the application

Keep Docker Desktop running. Use two PowerShell windows opened in the project folder.

Terminal 1 - Flask REST API

```
.\.venv\Scripts\Activate.ps1  
python run.py
```

Leave this terminal open. Verify the API in a browser: <http://127.0.0.1:5000/health>

Terminal 2 - Streamlit dashboard

```
.\.venv\Scripts\Activate.ps1  
streamlit run streamlit_app.py
```

Open <http://localhost:8501>. The sidebar provides Overview, Issues, and Vulnerabilities.

Quick acceptance check

Check	Expected result
API health	JSON shows status healthy and the company name
Overview	Four metrics and two charts load without an error
Create issue	New issue appears in the issue register
Update issue	Status or assignee change is saved
Create vulnerability	CVE, severity, CVSS, and asset are stored
Validation	Invalid or missing values show a clear error

4. Stop, restart, and troubleshoot

Stop the application

Press Ctrl+C in the Streamlit and Flask terminals. Then stop MySQL when it is no longer required:

```
docker compose down
```

Database records are retained in the Docker volume. Do not add the -v option unless the database should be permanently erased.

Normal restart

1. Start Docker Desktop.
2. Run `docker compose up -d mysql`.
3. Start `python run.py`.
4. Start `streamlit run streamlit_app.py`.

Common problems

Docker API / named pipe error

Open Docker Desktop and wait for Engine running, then retry `docker compose up -d mysql`.

Cannot connect to MySQL on 127.0.0.1

Confirm `docker compose ps` shows `northstar-mysql` as healthy. Then rerun the `init-db` command.

Port 5000 already in use

Close the previous Flask terminal or find and stop the existing development server.

Port 8501 already in use

Close the previous Streamlit terminal or run streamlit on another port.

Postman connection refused

Start Flask first and test `http://127.0.0.1:5000/health`.

Dark text/theme mismatch

Refresh with Ctrl+F5. Confirm `.streamlit/config.toml` exists in the project.

Support information

When requesting support, include the command that failed, the complete error text, the output of `docker compose ps`, and whether the health URL opens.